Supplementary Materials for

WaveletMask: Wavelet-Domain Mask-Guided Degradation Detection for Old Film Restoration

This document provides the supplementary figures and tables referenced from the main manuscript: complete diagnostic example grids (Figures S1–S3), the expert-level subjective quality evaluation (Tables S1–S2 and Figure S4), the spatial error-map analysis (Figure S5), and the probe visualization overview (Figure S6).

S1 Complete Diagnostic Example Grids

Figures S1–S3 provide the complete example grids for the three detector diagnostic probes reported in Appendix C of the main manuscript.

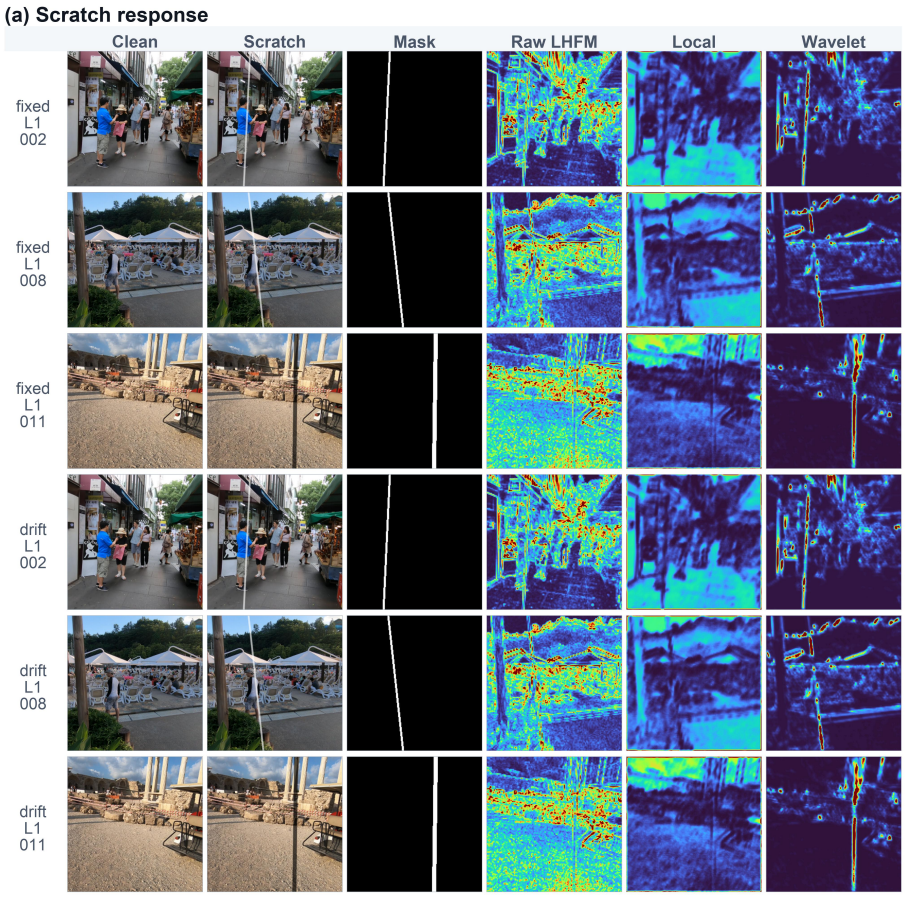


Figure S1: Complete persistent-scratch diagnostic examples. Each row compares clean frames, injected scratches, scratch masks, raw LHFM responses, learned local masks, and fused wavelet masks.

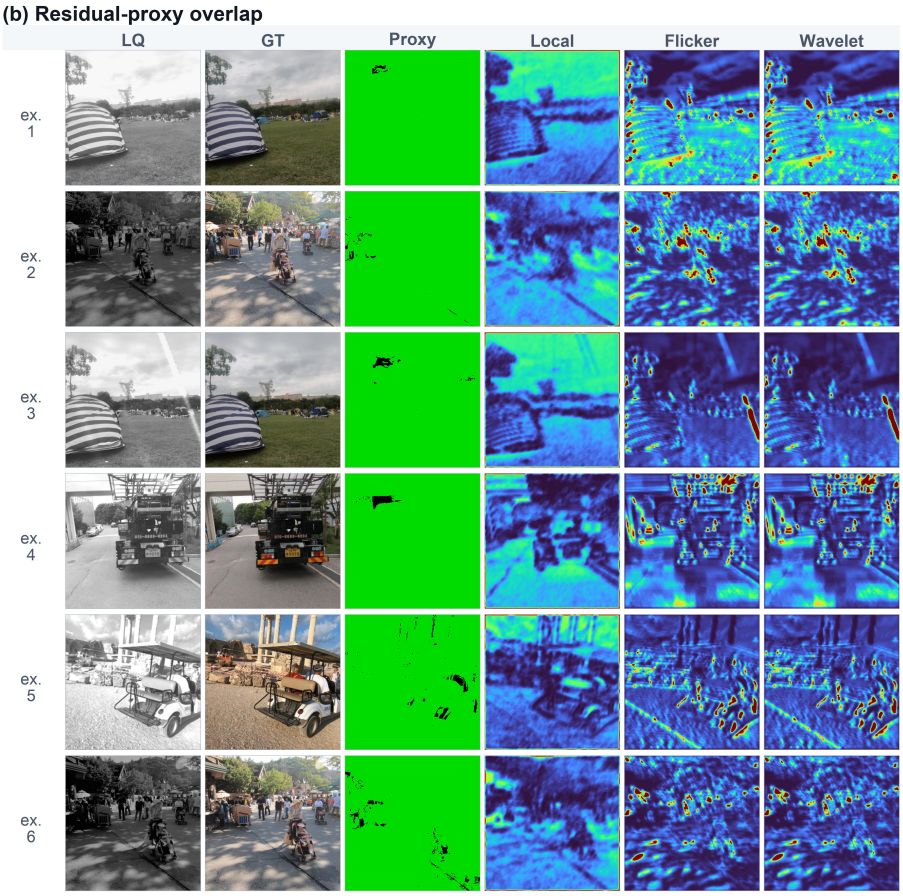


Figure S2: Complete residual-proxy overlap examples. Each row compares paired input–ground-truth residual proxies with learned local, flicker, and wavelet mask responses.

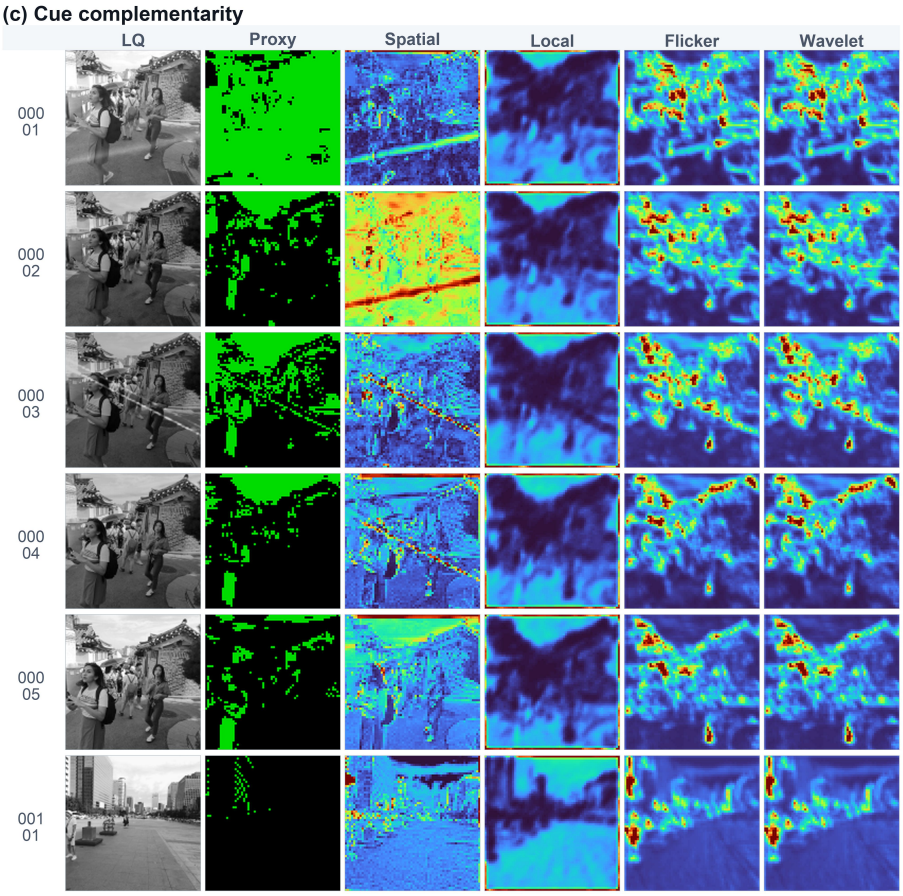


Figure S3: Complete cue-complementarity examples. Each row compares the flow-aligned spatial indicator with the local, flicker, and wavelet responses on the same frame.

S2 Expert-Level Subjective Quality Evaluation

To complement the content-sensitive no-reference metrics on the SRWOV real-world split, we conducted an internal expert visual-quality scoring with 34 domain-specialized internal raters. Each rater ranked four methods (WaveletMask, MambaOFR, RRTN, RTN) on 14 real old-film clips across three perceptual dimensions—degradation removal, style preservation, and temporal consistency—plus overall best visual experience. This internal panel provides an expert-level visual assessment while collecting only method-preference scores for the restored images. Presentation order was randomized per rater and per clip; raters were blind to method identity.

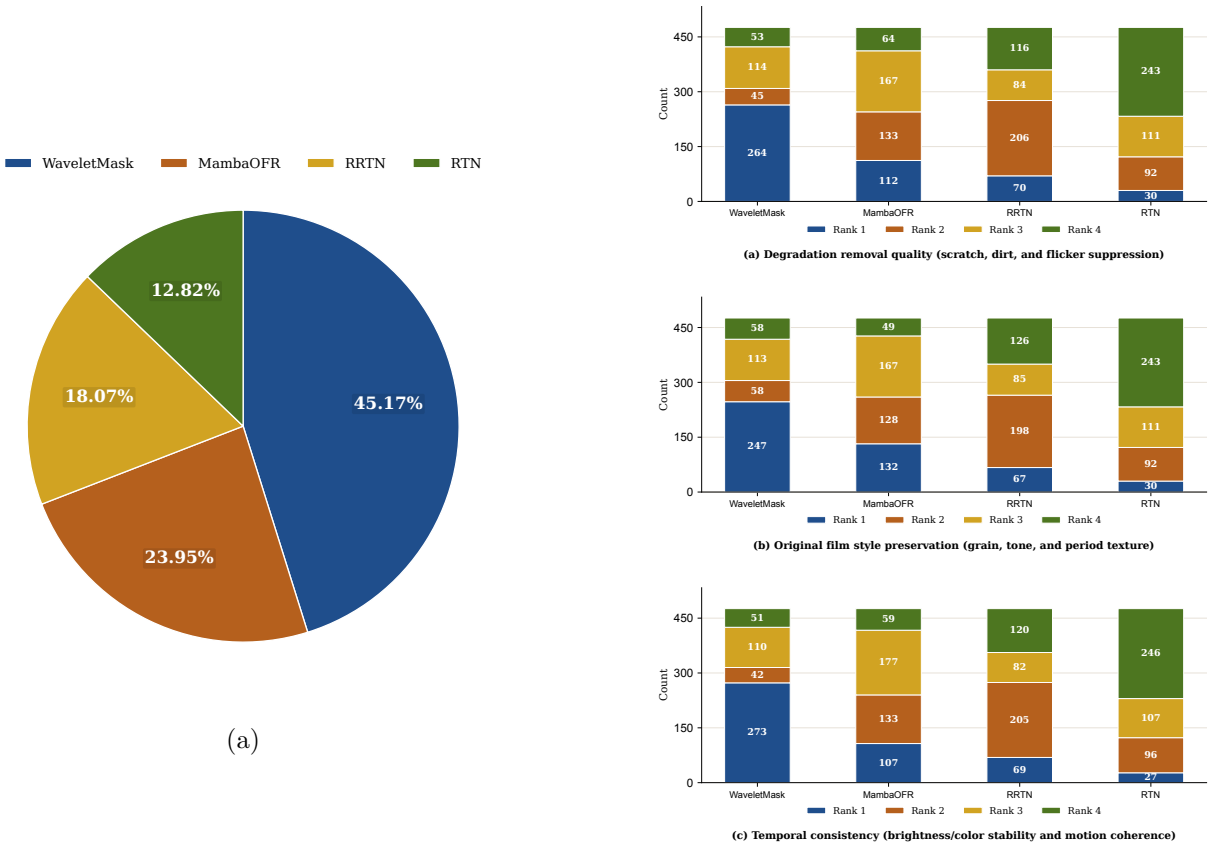
Table S1: Overall preference in the internal expert-level subjective evaluation (34 internal expert raters × 14 clips = 476 responses). Each response selects the method with the best visual experience. The best result is in red.

Method	Votes	Percent
WaveletMask	215	45.17%
MambaOFR	114	23.95%
RRTN	86	18.07%
RTN	61	12.82%

Table S2: Rank-1 counts by perceptual dimension (476 responses per dimension). The method receiving the most first-place votes per dimension is in red.

Dimension	WaveletMask	MambaOFR	RRTN	RTN
Degradation removal	264 (55.5%)	112 (23.5%)	70 (14.7%)	30 (6.3%)
Style preservation	247 (51.9%)	132 (27.7%)	67 (14.1%)	30 (6.3%)
Temporal consistency	273 (57.4%)	107 (22.5%)	69 (14.5%)	27 (5.7%)

Table S1 shows that WaveletMask receives 45.17% overall preference, outperforming the second-ranked MambaOFR (23.95%) by nearly 2× and RTN by 3.5×. Table S2 shows the same pattern across all three perceptual dimensions: WaveletMask ranks first in degradation removal (55.5%), style preservation (51.9%), and temporal consistency (57.4%). This is because LHFM suppresses local structural defects while GFM stabilizes low-frequency brightness fluctuation across frames. These preferences are statistically significant: the overall distribution deviates strongly from the uniform chance distribution ($\chi^2(3) = 115.1$, $p = 8.9 \times 10^{-25}$), and the WaveletMask share of 45.17% (Wilson 95% CI [40.8%, 49.7%]) lies significantly above the 25% chance level of the four-method ranking (one-sided exact binomial test, $p = 1.1 \times 10^{-21}$). The per-dimension rank-1 distributions are likewise significantly non-uniform (all $\chi^2(3) \geq 228.4$, $p < 10^{-48}$). These tests treat the 476 votes as independent draws and do not model rater-level clustering, because only aggregated vote exports are available. Figure S4 summarizes the overall preference distribution and the per-dimension ranking statistics.



(a)

(b)

Figure S4: Internal expert-level subjective evaluation results. (a) Overall preference distribution from the 34-rater internal study on 14 real old-film clips. (b) Rank-1 counts across three perceptual dimensions; WaveletMask achieves the highest first-place count in all dimensions.

S3 Spatial Error-Map Analysis

Figure S5 compares absolute error maps between the RTN and WaveletMask on selected synthetic frames with ground truth. The selected frames satisfy the recorded filter conditions used to generate the figure, including a PSNR gain above 1 dB, non-decreasing SSIM, and an improved-pixel ratio above 55%. The red improvement maps identify pixels where WaveletMask has lower absolute error than the baseline, showing that the gain is spatially concentrated around damaged regions in these examples.

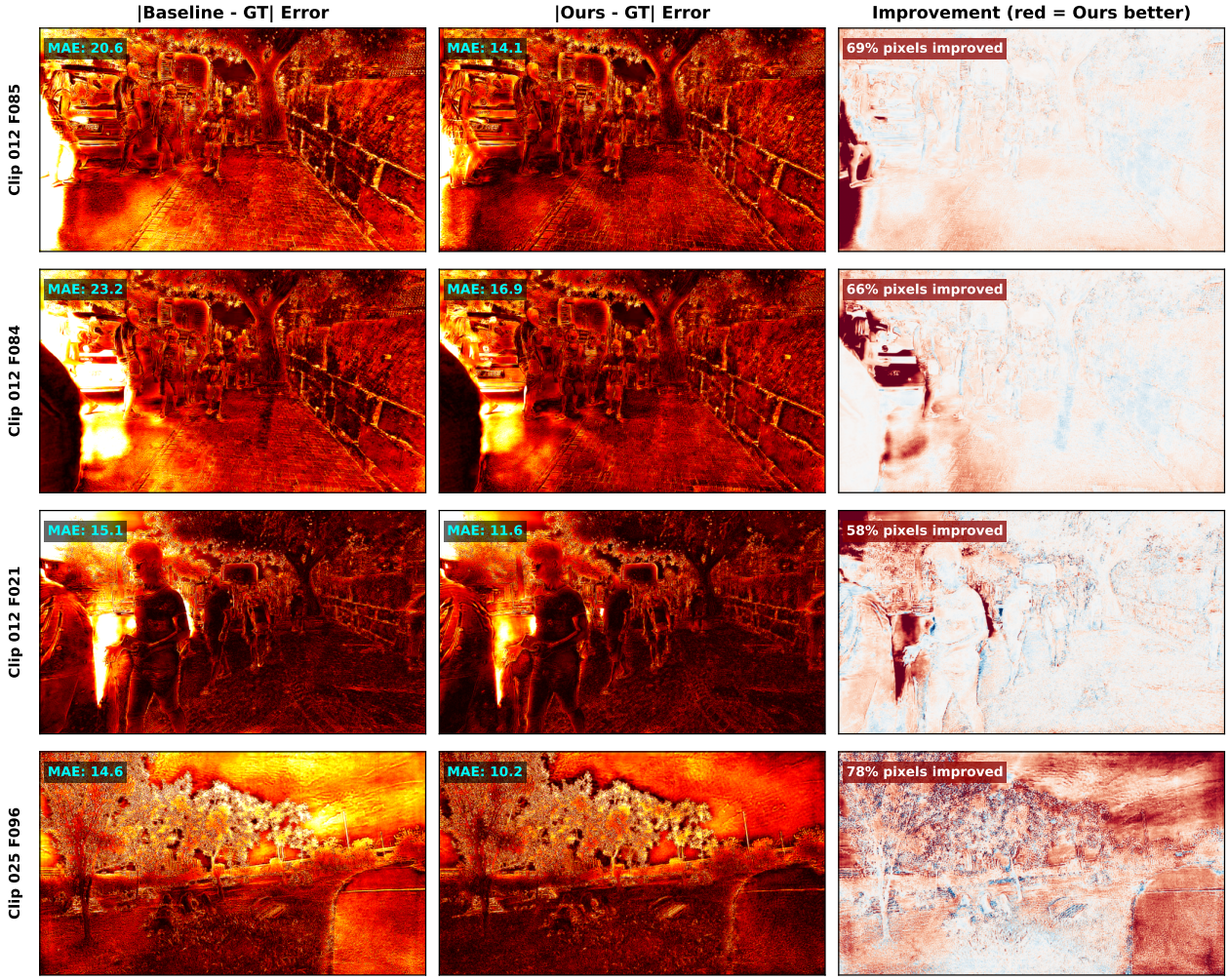


Figure S5: Error-map comparison between the RTN and WaveletMask on selected SRWOV synthetic frames. The first two columns report absolute error with respect to GT; the third column marks pixels where WaveletMask has lower error in red.

S4 Probe Visualization Overview

Figure S6 summarizes representative visual examples for the three detector probes; the persistent-scratch examples use single-frame ($L = 1$) injections and therefore highlight detector-response patterns, while the persistence-length trend is quantified in Appendix C of the main manuscript.

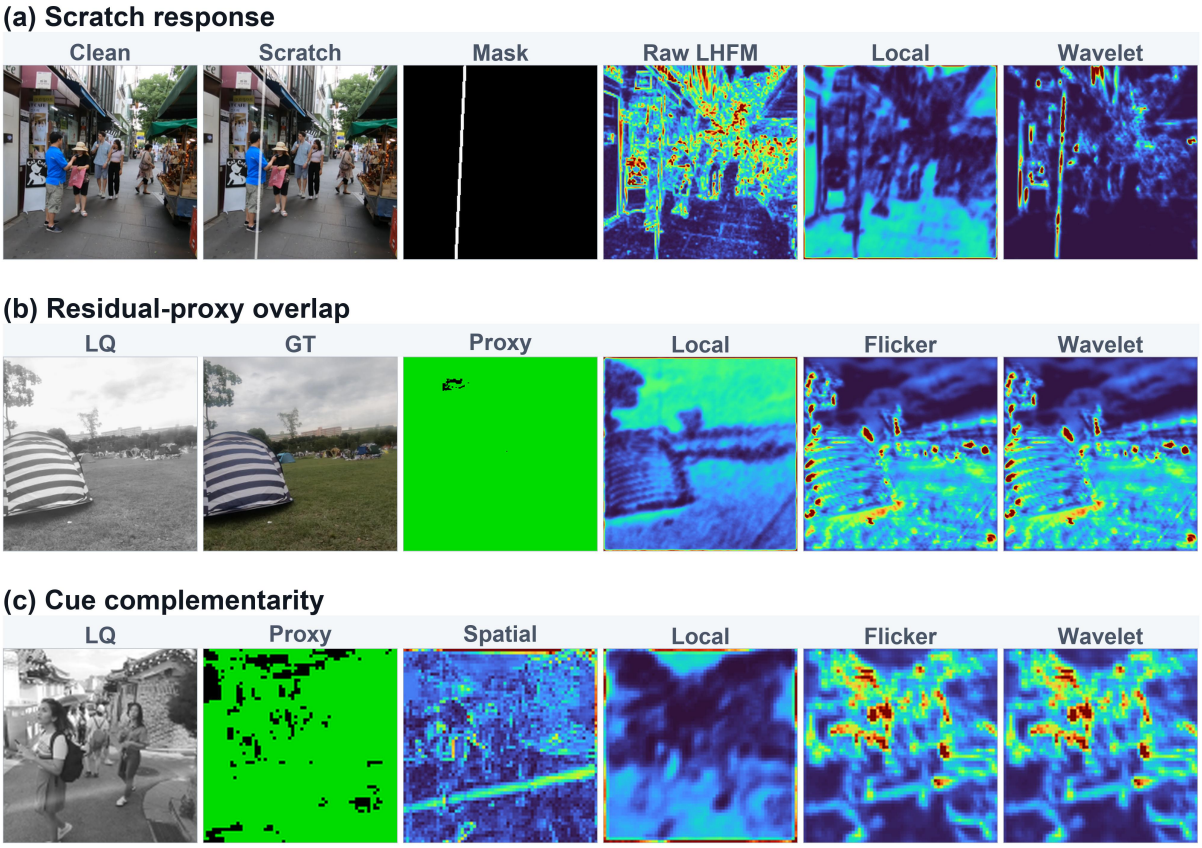


Figure S6: Visualization for the wavelet-mask probes. (a) Persistent-scratch examples compare clean frames, injected scratches, scratch masks, raw LHFM responses, learned local masks, and fused wavelet masks. (b) Residual-proxy examples compare paired input–ground-truth residual proxies with learned local, flicker, and wavelet mask responses. (c) Cue-complementarity examples compare the flow-aligned spatial indicator with the local, flicker, and wavelet responses on the same frame.